Stage Report on Linear Probe Experiments for Spatial Information Probing

1 Objective

The objective of this stage is to build and validate a *linear probe pipeline* for testing whether high-dimensional hidden representations encode *spatial information*.

To avoid the complexity and uncontrolled interference introduced by real data at the outset, this stage first focuses on experiments with dummy data. By using fully controllable synthetic data, we can verify whether the probe method is effective and observe its behavior under different data conditions.

Two sub-experiments were completed in this stage:

1. **dummy_linear_probe**: uses completely idealized and clean synthetic data to verify whether the probe pipeline works correctly.
2. **dummy_1_linear_probe**: constructs data that is closer to real-world scenarios (including room, object, tags, noise, etc.) to simulate the hidden representations encountered in realistic tasks.

2 Method

2.1 Core Idea: Linear Probe

The core idea of the experiment is as follows:

If hidden representations contain spatial information, then a simple linear model (Ridge regression) should be able to recover the relative spatial positions between objects from the hidden vectors.

Therefore, the experiment is formulated as a supervised learning problem:

- **Input**: a pair feature constructed from the hidden representations of two objects
- **Output**: the relative displacement between the two objects, namely

$$(dx, dz) = (x_j - x_i,\ z_j - z_i).$$

If the model achieves a high R^2 on the test set, this indicates that spatial information is linearly decodable from the hidden representation.

2.2 Dummy Data Construction

(1) `dummy_linear_probe`

The `dummy_linear_probe` experiment uses completely idealized synthetic data: object categories are sampled uniformly from a global vocabulary, position coordinates are generated independently and uniformly, and hidden vectors are composed only of a linear spatial mapping, category embeddings, and noise. This setting is used to verify whether the probe pipeline works correctly under clean conditions.

(2) `dummy_1_linear_probe`

The `dummy_1_linear_probe` experiment introduces structures that are closer to realistic scenarios based on the setup of `dummy_linear_probe`, including room-conditioned object co-occurrence, object-specific position priors, parent-child dependencies, relation tags, and hidden states jointly formed by spatial signal, category, room, scene, context mixing, and relation bias. In addition, spatial signal is restricted to only part of the hidden dimensions.

Compared with the first experiment, the second one has a much more complex data generation process and more closely resembles the statistical structure of realistic object-level hidden states. It therefore provides a stricter evaluation of the linear readability of spatial information by the probe.

2.3 Positive and Negative Controls

Two experimental conditions are considered:

Experiment	**Whether hidden contains spatial signal**
Positive Control	Yes
Negative Control	No

By comparing probe performance under these two conditions, we can determine whether the probe is truly using spatial information or merely exploiting other statistical biases.

2.4 Pair Feature Construction

For any pair of objects (i, j) in a scene, we construct the feature:

$$\text{feature} = [h_i;\ h_j;\ (h_j - h_i)].$$

The prediction target is:

$$(dx, dz) = (x_j - x_i,\ z_j - z_i).$$

The model is trained using Ridge Regression and evaluated on the test set with the following metrics:

- Mean Squared Error (MSE)
- R^2
- residual norm

Data splitting is performed at the *scene level* to avoid data leakage.

3 Experimental Results

3.1 `dummy_linear_probe` (Completely Idealized Data)

Test-set results are shown below:

Experiment	**Test R^2**
Positive	0.9997
Negative	-0.1059

These results show that:

- when hidden representations contain spatial information, the linear probe can almost perfectly recover relative positions;
- when hidden representations do not contain spatial information, the model fails to predict the target ($R^2 < 0$).

This indicates that the probe pipeline is **effective and correct** under ideal conditions. The role of this experiment is to verify that the method itself is feasible.

3.2 `dummy_1_linear_probe` (Data Closer to Realistic Scenarios)

Test-set results are shown below:

Experiment	**Test R^2**
Positive	0.9986
Negative	0.6353

Detailed test metrics are as follows.

Positive Control (with spatial signal)

- Test $R^2 = 0.9986$
- MSE $= 0.0062$
- Mean residual norm ≈ 0.0996

Negative Control (without spatial signal)

- Test $R^2 = 0.6353$
- MSE $= 1.8611$
- Mean residual norm ≈ 1.6750

4 Result Analysis

4.1 Positive Control

When hidden representations contain spatial signal:

- the test-set performance reaches $R^2 \approx 0.9986$;
- relative positions can be recovered almost perfectly.

This indicates that when hidden representations indeed encode spatial information, a linear probe can successfully decode that spatial information.

This verifies that the linear probe is sensitive and effective for spatial information.

4.2 Negative Control (Important Observation)

Unlike in the first dummy experiment, in `dummy_1_linear_probe` the Negative Control still achieves:

$$R^2 = 0.6353,$$

which is far from 0 or a negative value.

This reveals an important phenomenon:

Even when hidden representations do not contain explicit spatial signal, the model can still partially predict spatial relations by exploiting semantic information such as room, object, and tags.

Possible reasons include:

1. Some objects have statistical position preferences within a room.
2. Tags such as `left`, `right`, and `near` themselves contain weak spatial information.
3. The combination of room embeddings and object embeddings forms spatial priors.

Therefore, the performance of the probe may contain at least two sources:

1. predictability induced by semantic or scene-level statistical structure;
2. genuinely additional and finer-grained geometric spatial information.

5 Roles of the Two Sub-Experiments

Experiment	**Role**
`dummy_linear_probe`	Verify that the probe method itself is correct
`dummy_1_linear_probe`	Simulate realistic hidden states and observe probe behavior under mixed informat

The experimental pipeline has therefore been validated and is ready to move to the real-data stage.

6 Current-Stage Conclusions

This stage completed the construction of the linear probe pipeline and verified the following conclusions:

1. Linear probe can effectively detect whether hidden representations contain spatial information.
2. In idealized data:
   - with spatial signal: $R^2 \approx 1$;
   - without spatial signal: $R^2 \approx 0$ or negative.
3. In data closer to realistic scenarios:
   - even without explicit spatial signal, the probe can still partially predict spatial relations using semantic or statistical information ($R^2 \approx 0.63$).
4. Therefore, when probe methods are used in real tasks, one should be cautious:

- probe success does not necessarily imply that the model has learned a geometric spatial representation;
- it may instead reflect exploitation of statistical biases from object, room, or tag information.

7 Next Stage: Real-World Data Experiments

The next stage will transition the experiments from dummy data to a small amount of real-world scene data. The goal of this stage is to construct a *minimum viable pipeline* for real data and to run linear probe experiments on a small real-world dataset for preliminary analysis.

7.1 Real-World Spatial Data Exploration and Minimal Dataset Construction

The first step is to explore the data structure of AI2-THOR scenes and construct a minimal viable dataset to run through the full pipeline, from real spatial data to language model representations and then to linear probing.

The information to be extracted initially includes:

- object category in each scene;
- spatial coordinates of each object, (x, z);
- scene or floor-plan identifier (used for train/test split).

Based on object-level coordinates, we will first construct the most basic pairwise spatial supervision signal, namely the relative displacement vector between objects:

$$(\Delta x, \Delta z) = (x_j - x_i,\ z_j - z_i).$$

This continuous variable contains both distance and direction information and can serve as a unified spatial supervision signal. It can also be further used to derive discrete spatial relations such as *left/right*, *front/back*, and *near/far*.

7.2 Multi-Object Text Generation and Hidden Representation Extraction

After obtaining objects and basic spatial relations in each scene, we will use simple rule-based templates to generate multi-object spatial description texts. These texts will then be fed into a frozen language model to extract hidden representations.

For each scene, we will generate textual descriptions that contain multiple objects and their spatial relations. Hidden states will be extracted from multiple transformer layers of the language model. By pooling over the token span corresponding to each object, we obtain object-level representation vectors.

Based on these object-level representations, and following the dummy experiments, we construct pair features for each object pair (i, j), which will serve as the input features for the probe.

7.3 Small-Scale Linear Probe and Preliminary Analysis

We will train linear probes on a small amount of real-world data, using object-pair hidden representations to predict relative displacement $(\Delta x, \Delta z)$. The following metrics will be recorded:

- Mean Squared Error (MSE);
- R^2 (coefficient of determination).

The main goal of this stage is to observe whether spatial displacement is linearly decodable from real language model hidden representations to some extent.

In this experiment, MSE and R^2 are used as the primary evaluation metrics for the linear probe. MSE measures the magnitude of the error between the predicted relative displacement and the ground-truth displacement, while R^2 measures the proportion of variance in spatial displacement that can be explained by the model.

It is important to note that a high R^2 indicates that spatial information can be linearly recovered from the representations, but it does not necessarily imply that the representation space itself has a geometric structure consistent with real space. As observed in the dummy experiments, probe performance may be supported not only by true geometric information but also by statistical or semantic regularities correlated with spatial relations.

Therefore, in subsequent analysis, we will also examine the correlation between distances in representation space and distances in real space, as well as geometric alignment analysis, in order to distinguish between superficial predictability and true geometric structure representation.

Appendix

A `dummy_1_linear_probe` Data Construction Details

The goal of `dummy_1_linear_probe` is to preserve the assumption that hidden representations still contain linearly decodable spatial information, while making the statistical structure of the synthetic data closer to realistic indoor scenes and realistic model hidden states.

Compared with the first experiment, this setting no longer uses completely independent, uniformly sampled objects and coordinates. Instead, it explicitly introduces room types, object co-occurrence priors, position priors, relation tags, and contextual interference terms.

In this experiment, each object in a scene is no longer described simply by "category + coordinate + hidden", but instead has richer attributes including:

- **room**: room type, such as `kitchen`, `living_room`, and `office`;
- **object category**: object type, such as `cup`, `chair`, and `lamp`;
- **position** (x, z): two-dimensional position;
- **tags**: semantic relation labels inferred from scene relations, such as `near`, `left`, `right`, `on`, and `in`;
- **hidden vector**: the final high-dimensional representation.

(1) Room-Conditioned Object Sampling

Unlike the first experiment, `dummy_1_linear_probe` does not directly sample objects from a global object vocabulary. Instead, it first samples a room type for each scene, and then samples the actual objects from the candidate object set associated with that room. For example:

- a `kitchen` is more likely to contain `fridge`, `microwave`, `plate`, and `cup`;
- a `living_room` is more likely to contain `sofa` and `lamp`;
- an `office` is more likely to contain `chair`, `table`, and `cabinet`.

Thus, the object distribution within a scene has a more realistic co-occurrence structure. This differs from the first experiment, where all object categories appear independently and with equal probability.

(2) Position Sampling with Priors and Dependencies

For position generation, this experiment also no longer samples all object positions independently and uniformly from the full space. Instead, each object category is assigned a position-prior center. For example, `fridge`, `table`, `sofa`, and `lamp` are each more likely to appear in certain regions.

In addition, simple parent-child or receptacle-like dependencies are introduced:

- if a `table` already exists in the scene, then `plate` and `cup` are more likely to be distributed near the table;
- `apple` is sometimes likely to be near a `fridge`; otherwise, if a `table` exists, it is also more likely to appear near the table.

Therefore, object positions in this experiment are no longer mutually independent, but instead reflect more realistic indoor spatial structures. In contrast, the first experiment used fully independent position sampling without any category prior or relation dependency.

(3) Inference of Relation Tags

In `dummy_1_linear_probe`, each object is also assigned several relation tags according to its relative spatial relations with other objects. For example:

- if an object is attached to a `table`, it may receive the tag `on`;
- if its parent is a `fridge` or `cabinet`, it may receive the tag `in`;
- if it is very close to another object, it may receive the tag `near`;
- if another object is clearly on its left or right side, it may receive the tag `left` or `right`.

These tags are not directly used as supervision targets. Instead, they further participate in the construction of hidden vectors, thereby injecting relation-semantic bias into hidden representations. This setting is absent from the first experiment.

(4) More Complex Hidden Construction

This is the most important difference between the second experiment and the first one.

In the first experiment, hidden vectors are composed only of "spatial term + category term + noise", whereas in `dummy_1_linear_probe`, the hidden vector is formed by the sum of multiple sources:

$$h = \underbrace{W_{\text{space}} \cdot (x, z)}_{\text{spatial signal}} + \underbrace{E_{\text{category}}}_{\text{category embedding}} + \underbrace{E_{\text{room}}}_{\text{room-type embedding}} + \underbrace{E_{\text{scene}}}_{\text{scene-level context}} + \underbrace{E_{\text{context-mix}}}_{\text{cross-object semantic mixing}} + \underbrace{E_{\text{tag}}}_{\text{relation-tag b}}$$

The meanings of these components are as follows:

- **spatial signal**: projects (x, z) into hidden space through a linear mapping;
- **category embedding**: represents object-category semantics;
- **room embedding**: represents the shared context of the room type to which the scene belongs;
- **scene-level context**: a randomly sampled shared vector for each scene;
- **cross-object semantic mixing**: formed by the mean of all object-category embeddings in the current scene, used to simulate the effect that object-level hidden states are influenced by global scene context;
- **relation-tag bias**: semantic bias induced by tags such as `near`, `left`, `right`, `on`, and `in`;
- **noise**: independent Gaussian noise.

In addition, this experiment further specifies that spatial information is not written into all hidden dimensions. Instead, it is written only into a subset of hidden dimensions, controlled by a `space_mask`, so that the spatial component enters only certain subspaces. This setting is intended to simulate the realistic case where information is often concentrated in local blocks of dimensions within the model representation. In contrast, in the first experiment, spatial information is directly mapped into all hidden dimensions.

(5) Supervision Target and Probe Task

Although the data construction becomes substantially more complex, the probe task itself remains identical to that of the first experiment: for all ordered object pairs within the same scene, we construct the pair feature

$$[h_i;\ h_j;\ h_j - h_i],$$

and use the true relative displacement (dx, dz) as the supervision target to train a Ridge regression probe for recovering relative spatial relations.

Therefore, the second experiment does not change the probe task itself. Instead, while keeping the task fixed, it makes the hidden generation mechanism more complex, thereby providing a stricter test of the effectiveness of the probe.